

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	139893	46.015769
1	[Git, Change, Create]	45414	14.938275
2	[Git, Change, Delete]	22797	7.498742
3	[Java, ByteCode, Member, Method]	13818	4.545230
4	[Java, ByteCode, Parameter]	13459	4.427142
5	[Git, Commit]	11609	3.818612
6	[File, Git]	8332	2.740690
7	[Java, ByteCode, Bound, ParameterizedType]	7323	2.408794
8	[Java, ByteCode, Bound]	7320	2.407808
9	[Git, Change, Rename]	7018	2.308469
10	[Java, ByteCode, Member, Field]	3724	1.224956
11	[Java, Value, ByteCode, Annotation]	2958	0.972991
12	[Java, ByteCode, Bound, WildcardType]	2950	0.970360
13	[Java, ByteCode, Member, Method, Constructor]	2180	0.717079
14	[Xml, Element]	2162	0.711158
15	[Xml, Text]	1450	0.476956
16	[Java, ByteCode, TypeVariable, Bound]	1112	0.365776
17	[Java, ByteCode, Member, Method, Lambda]	986	0.324330
18	[Type, File, Java, ByteCode, ResolvedDuplicate...	903	0.297029
19	[Type, File, Java, Class, ByteCode]	869	0.285845
20	[Json, Key]	750	0.246702
21	[Value, Json, Scalar]	734	0.241439
22	[Java, Value, ByteCode, Primitive]	686	0.225650
23	[Type, File, Java, ByteCode, JavaType]	659	0.216768
24	[Java, GenericDeclaration, ByteCode, Member, M...	579	0.190454
25	[Type, File, Java, ByteCode, ExternalType]	419	0.137824
26	[Git, Change, Copy]	298	0.098023
27	[Author, Git, Person]	297	0.097694
28	[Value, Array]	294	0.096707
29	[Committer, Git, Person]	243	0.079931

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

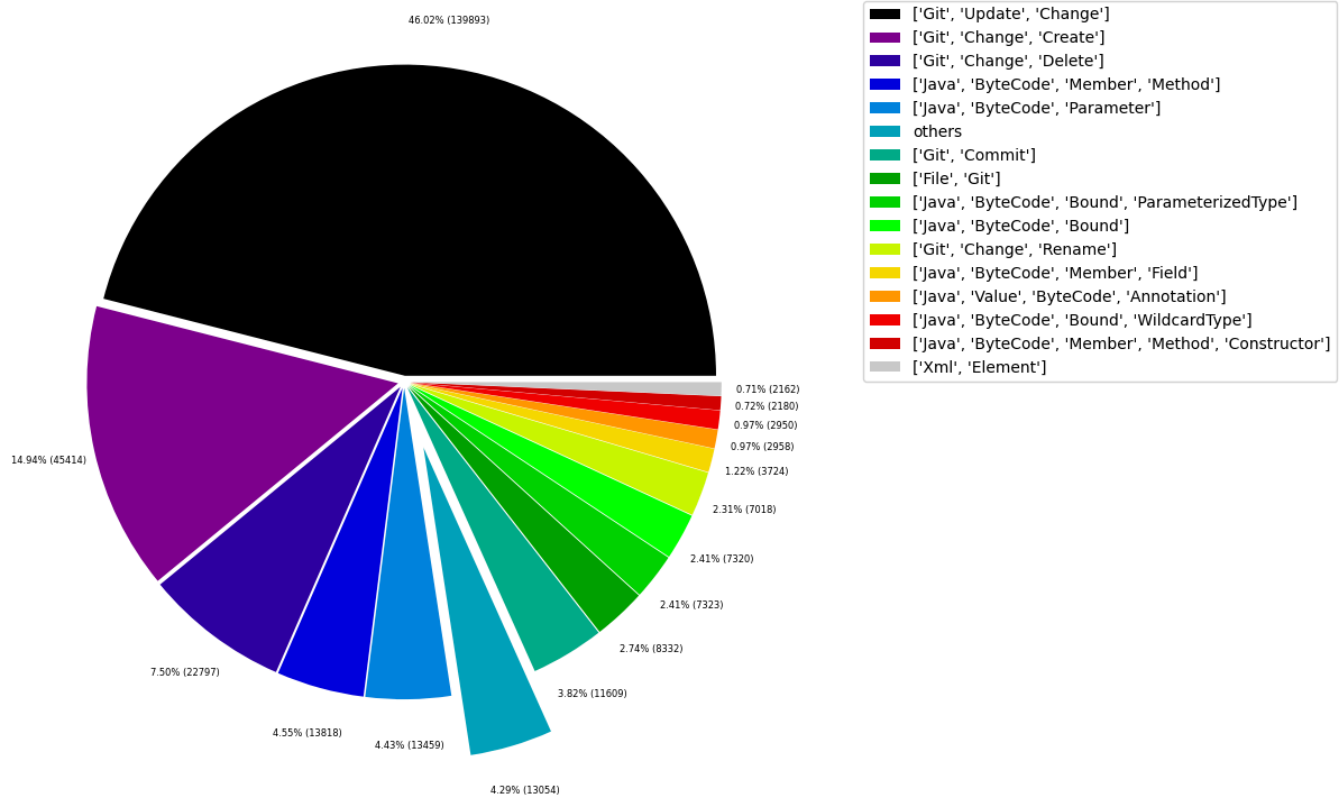


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000329
1	[Git, Branch]	1	0.000329
2	[Repository, File, Git]	1	0.000329
3	[File, Json]	2	0.000658
4	[File]	3	0.000987
5	[Java, GenericDeclaration, ByteCode, Member, M...	4	0.001316
6	[Maven, Exclusion]	5	0.001645
7	[Value, Array, Json]	6	0.001974
8	[File, Maven, Xml, Pom, Document]	9	0.002960
9	[Type, File, Java, ByteCode, Void]	9	0.002960
10	[Java, ManifestSection]	9	0.002960
11	[File, Artifact, Jar, Archive, Zip, Java]	9	0.002960
12	[File, Java, Manifest]	9	0.002960
13	[File, Java, ServiceLoader]	11	0.003618
14	[File, Java, Properties]	12	0.003947
15	[Maven, ExecutionGoal]	16	0.005263
16	[Maven, PluginExecution]	16	0.005263
17	[Xml, Attribute]	18	0.005921
18	[jQAssistant, Rule, Concept]	19	0.006250
19	[Type, File, Java, ByteCode, Throwable, Extern...	21	0.006908
20	[Maven, Plugin]	21	0.006908
21	[Maven, Configuration]	21	0.006908
22	[Type, File, Java, ByteCode, Throwable, Resolv...	22	0.007237
23	[Type, File, Java, ByteCode, Enum]	30	0.009868
24	[Type, File, Java, ByteCode, PrimitiveType]	31	0.010197
25	[Xml, Namespace]	36	0.011842
26	[Type, File, Java, ByteCode, Annotation]	44	0.014473
27	[File, Directory]	45	0.014802
28	[Type, File, Java, Class, ByteCode, Throwable]	56	0.018420
29	[Type, File, Java, GenericDeclaration, ByteCod...	85	0.027960

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

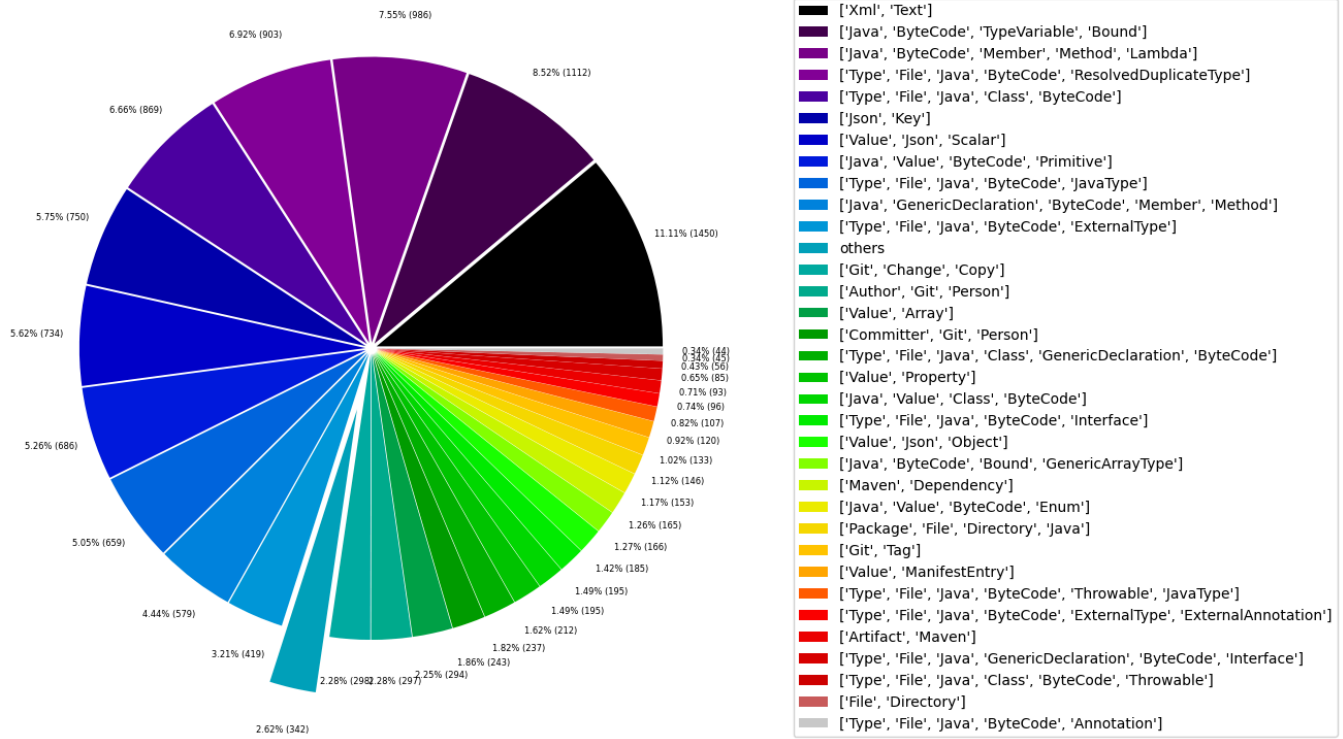


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

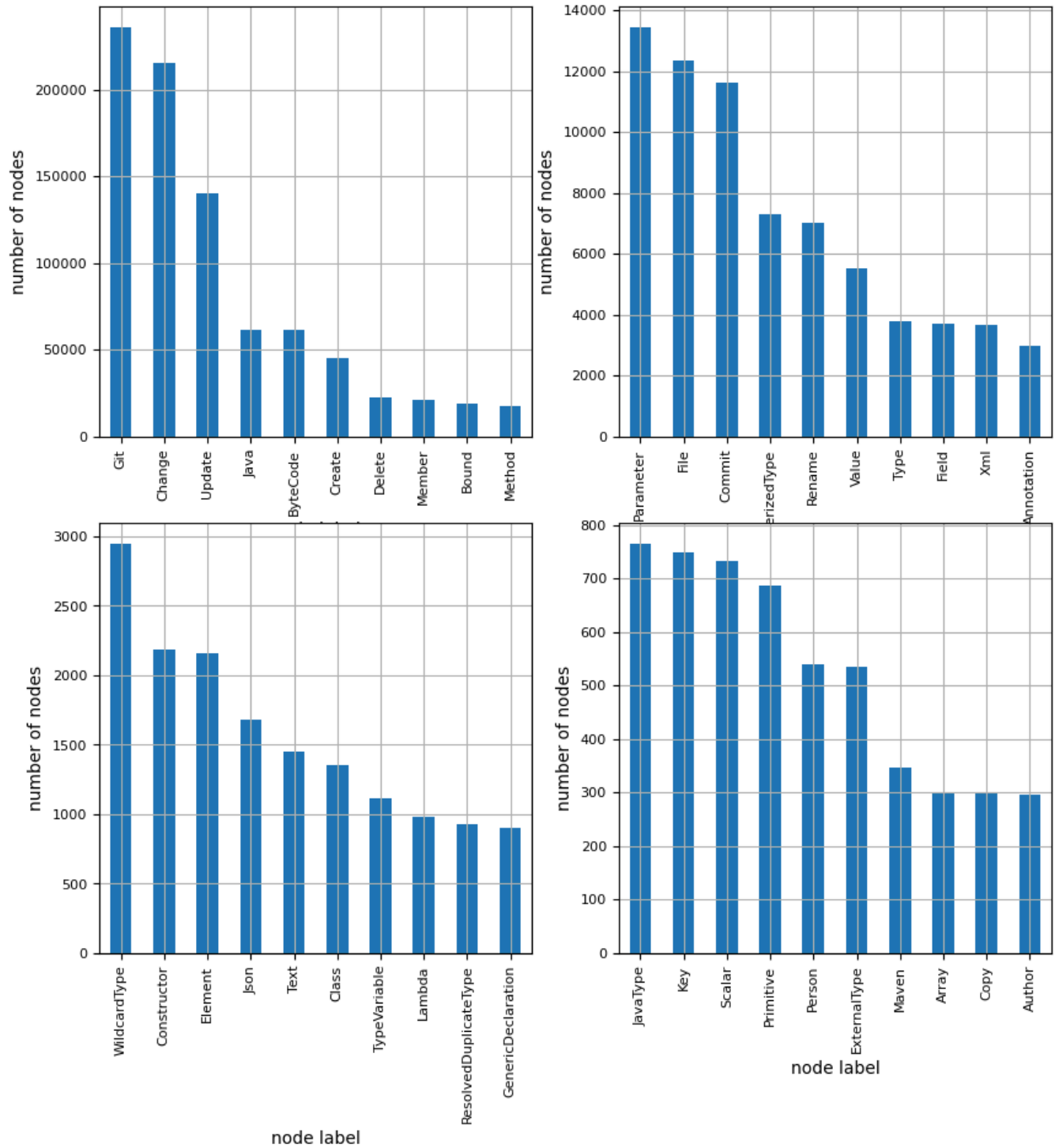
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	236036	77.640612
1	Change	215420	70.859278
2	Update	139893	46.015769
3	Java	61592	20.259793
4	ByteCode	61396	20.195322
5	Create	45414	14.938275
6	Delete	22797	7.498742
7	Member	21291	7.003365
8	Bound	18871	6.207341
9	Method	17567	5.778409
10	Parameter	13459	4.427142
11	File	12362	4.066300
12	Commit	11609	3.818612
13	ParameterizedType	7323	2.408794
14	Rename	7018	2.308469
15	Value	5543	1.823289
16	Type	3783	1.244363
17	Field	3724	1.224956
18	Xml	3675	1.208838
19	Annotation	3002	0.987464
20	WildcardType	2950	0.970360
21	Constructor	2184	0.718395
22	Element	2162	0.711158
23	Json	1677	0.551625
24	Text	1450	0.476956
25	Class	1357	0.446365
26	TypeVariable	1112	0.365776
27	Lambda	986	0.324330
28	ResolvedDuplicateType	925	0.304265
29	GenericDeclaration	905	0.297687
30	JavaType	766	0.251965
31	Key	750	0.246702
32	Scalar	734	0.241439
33	Primitive	686	0.225650
34	Person	540	0.177625
35	ExternalType	536	0.176309
36	Maven	346	0.113812
37	Array	300	0.098681
38	Copy	298	0.098023
39	Author	297	0.097694

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 968664

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	215420	22.238877
1	MODIFIES	215420	22.238877
2	UPDATES	139893	14.441850
3	CREATES	52730	5.443580
4	INVOKES	37644	3.886177
5	DELETES	29815	3.077951
6	CHANGED_TOGETHER_WITH	26210	2.705789
7	COMMITTED	23218	2.396910
8	DEPENDS_ON	22918	2.365939
9	OF_TYPE	22230	2.294913
10	DECLARES	21756	2.245980
11	OF_RAW_TYPE	17483	1.804857
12	HAS	14616	1.508882
13	HAS_PARENT	14206	1.466556
14	RETURNS	13084	1.350726
15	HAS_COMMIT	11609	1.198455
16	READS	9705	1.001895
17	HAS_ACTUAL_TYPE_ARGUMENT	8459	0.873265
18	HAS_FILE	8332	0.860154
19	RENAMES	7018	0.724503
20	OF_GENERIC_TYPE	6045	0.624055
21	RESOLVES_TO	4311	0.445046
22	SIMILAR	4133	0.426670
23	WRITES	4053	0.418411
24	CONTAINS	4010	0.413972
25	HAS_NEW_NAME	3662	0.378046
26	RETURNS_GENERIC	3616	0.373298
27	ANNOTATED_BY	2946	0.304130
28	REQUIRES	2267	0.234034
29	HAS_FIRST_CHILD	2162	0.223194

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

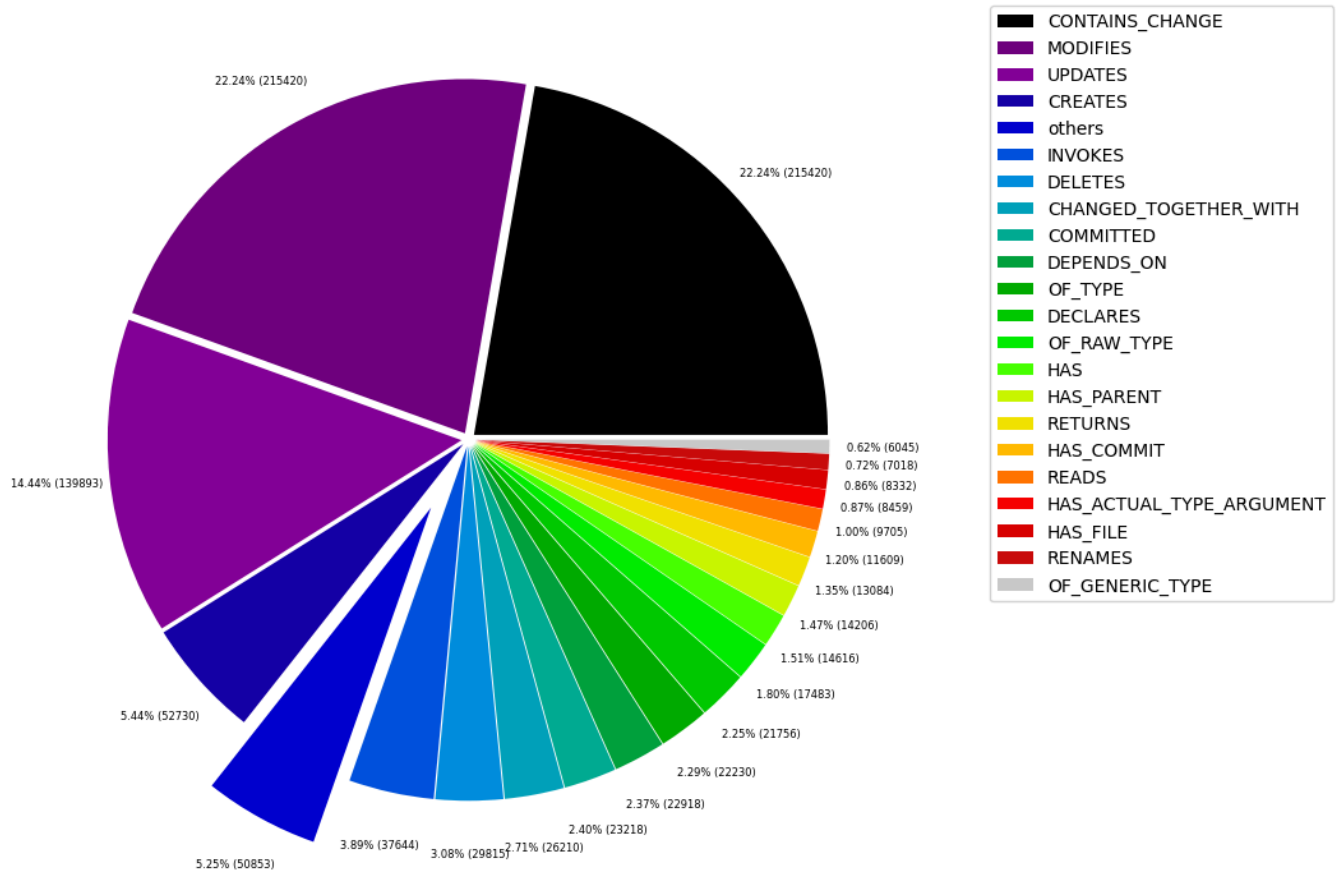


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

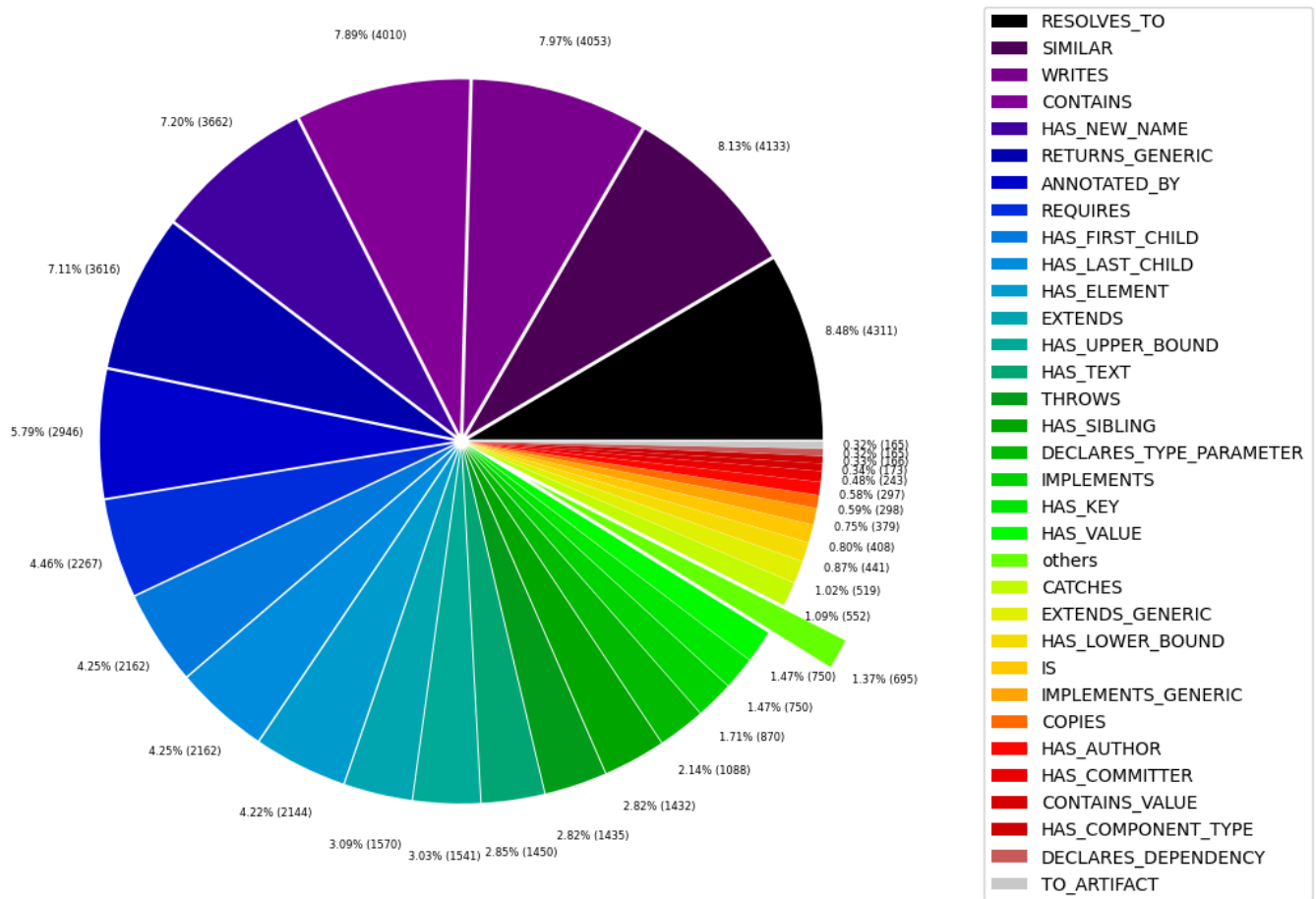
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000103
1	HAS_BRANCH	1	0.000103
2	HAS_HEAD	2	0.000206
3	THROWS_GENERIC	5	0.000516
4	EXCLUDES	5	0.000516
5	DESCRIBES	9	0.000929
6	HAS_ROOT_ELEMENT	10	0.001032
7	HAS_GOAL	16	0.001652
8	HAS_EXECUTION	16	0.001652
9	OF_NAMESPACE	18	0.001858
10	HAS_ATTRIBUTE	18	0.001858
11	INCLUDES_CONCEPT	19	0.001961
12	USES_PLUGIN	21	0.002168
13	IS_ARTIFACT	21	0.002168
14	HAS_CONFIGURATION	21	0.002168
15	REQUIRES_TYPE_PARAMETER	24	0.002478
16	REQUIRES_CONCEPT	28	0.002891
17	DECLARES_NAMESPACE	36	0.003716
18	HAS_DEFAULT	39	0.004026
19	COPY_OF	119	0.012285
20	HAS_TAG	133	0.013730
21	ON_COMMIT	133	0.013730
22	DECLARES_DEPENDENCY	165	0.017034
23	TO_ARTIFACT	165	0.017034
24	HAS_COMPONENT_TYPE	166	0.017137
25	CONTAINS_VALUE	173	0.017860
26	HAS_COMMITTER	243	0.025086
27	HAS_AUTHOR	297	0.030661
28	COPIES	298	0.030764
29	IMPLEMENTS_GENERIC	379	0.039126

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numbe
0	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	139893	11609	
1	[Git, Update, Change]	MODIFIES	[File, Git]	139893	139893	
2	[Git, Update, Change]	UPDATES	[File, Git]	139893	139893	
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	45414	11609	
4	[Git, Change, Create]	CREATES	[File, Git]	45414	45414	
5	[Git, Change, Create]	MODIFIES	[File, Git]	45414	45414	
6	[Java, ByteCode, Member, Method]	INVOKES	[Java, ByteCode, Member, Method]	22985	13717	
7	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	22797	11609	
8	[Git, Change, Delete]	MODIFIES	[File, Git]	22797	22797	
9	[Git, Change, Delete]	DELETES	[File, Git]	22797	22797	
10	[File, Git]	CHANGED_TOGETHER_WITH	[File, Git]	20075	8332	
11	[Git, Commit]	HAS_PARENT	[Git, Commit]	14197	11609	
12	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	11609	1	
13	[Author, Git, Person]	COMMITTED	[Git, Commit]	11609	297	
14	[Committer, Git, Person]	COMMITTED	[Git, Commit]	11609	243	
15	[Java, ByteCode, Member, Method]	READS	[Java, ByteCode, Member, Field]	8730	13717	
16	[Java, ByteCode, Member, Method]	HAS	[Java, ByteCode, Parameter]	8602	13717	
17	[Repository, File, Git]	HAS_FILE	[File, Git]	8332	1	
18	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	7018	11609	
19	[Git, Change, Rename]	MODIFIES	[File, Git]	7018	7018	
20	[Git, Change, Rename]	CREATES	[File, Git]	7018	7018	
21	[Git, Change, Rename]	RENAMES	[File, Git]	7018	7018	
22	[Git, Change, Rename]	DELETES	[File, Git]	7018	7018	
23	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, javaType]	6243	13459	
24	[File, Git]	HAS_NEW_NAME	[File, Git]	3662	8332	
25	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, javaType]	3501	7320	
26	[Java, ByteCode, Bound, ParameterizedType]	OF_RAW_TYPE	[Type, File, Java, ByteCode, javaType]	3151	7323	
27	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, WildcardType]	2950	7323	
28	[Java, ByteCode, Parameter]	OF_GENERIC_TYPE	[Java, ByteCode, Bound, ParameterizedType]	2712	13459	
29	[Java, ByteCode, Member, Method, Constructor]	WRITES	[Java, ByteCode, Member, Field]	2585	2180	

Graph Density

```
total_number_of_nodes (vertices): 304011
total_number_of_relationships (edges): 968664
-> total directed graph density: 1.0480837616643907e-05
-> total directed graph density in percent: 0.0010480837616643906
```